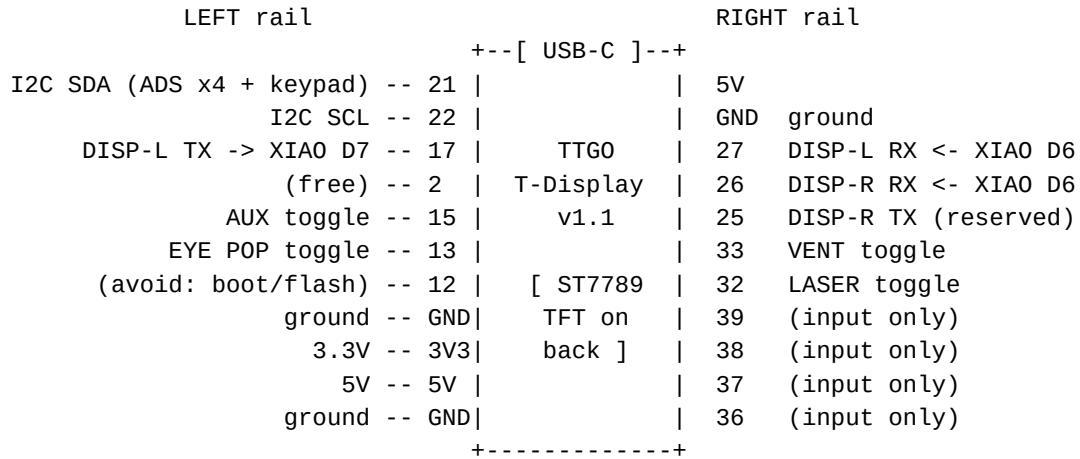


j4_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)



on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO00,
GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, ADS_03 @ 0x4A,
ADS_04 @ 0x4B, PCF8574 keypad @ 0x20
toggles: INPUT_PULLUP, switch closes to GND (ON = LOW)

GPIO	Function
21	I2C SDA (ADS1115 x4, PCF8574 keypad)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, reserved)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
32	LASER toggle (INPUT_PULLUP, switch closes to GND)
33	VENT toggle (INPUT_PULLUP, switch closes to GND)
13	EYE POP toggle (INPUT_PULLUP, switch closes to GND)
15	AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in; data / MAC / connection status)

j4_receiver -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)

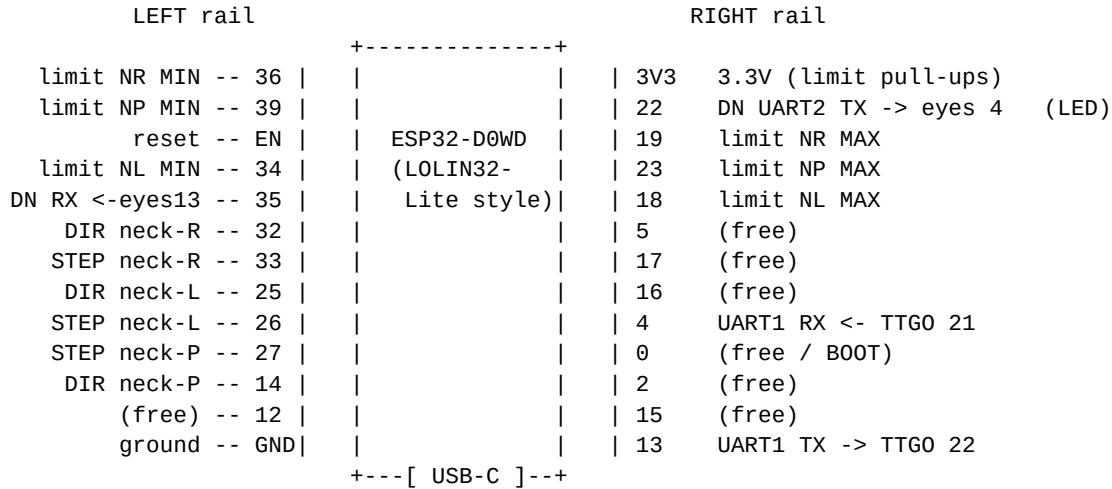
LEFT rail		RIGHT rail
	+--[USB-C]--+	
UART1 TX -> j4_stepper_neck 4 -- 21		5V
UART1 RX <- j4_stepper_neck 13 -- 22		GND ground
UART2 TX -> Teensy 25 (RX6) -- 17	TTGO	27 (free)
UART2 RX <- Teensy 24 (TX6) -- 2	T-Display	26 (free)
(free) -- 15	v1.1	25 WS2812B data
(free) -- 13		33 I2C SCL (PCA9685)
(free) -- 12	[ST7789	32 I2C SDA (PCA9685)
ground -- GND	TFT on	39 (input only)
3.3V -- 3V3	back]	38 (input only)
5V -- 5V		37 (input only)
ground -- GND		36 (input only)
	+-----+	

on-board (no header pin): GPIO34 = battery sense, B00T = GPIO00,
GPIO35 = screen mode button, TFT on 4/5/16/18/19/23

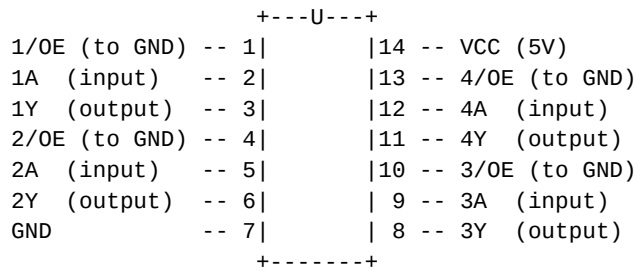
GPIO	Function
2	UART2 RX from Teensy (Teensy Serial6 TX, pin 24)
17	UART2 TX to Teensy (Teensy Serial6 RX, pin 25)
21	UART1 TX to j4_stepper_neck (250000 baud)
22	UART1 RX from j4_stepper_neck (status + EY: return)
25	WS2812B LED strip data (through ~330 ohm series resistor)
32	I2C SDA (PCA9685)
33	I2C SCL (PCA9685)
34	Battery voltage sense (ADC, read-only)
35	Screen mode button

j4_stepper_neck -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM



SN74AHCT125N x2 (DIP-14, top view, notch at the top)



U1: 1A<-26 1Y->NL PUL+ ; 2A<-25 2Y->NL DIR+ ; 3A<-33 3Y->NR PUL+ ; 4A<-32 4Y->NR DIR+
U2: 1A<-27 1Y->NP PUL+ ; 2A<-14 2Y->NP DIR+ ; 3A,4A->GND (spare) ; 3Y,4Y n/c
GPIO 21 is NOT present on this board, which is why the downstream TX is on 22.
Neck STEP/DIR pins pass through the 74AHCT125 buffers (3.3V -> 5V) into the DM556TE PUL+/DIR+ inputs (PUL-/DIR- to GND). 100nF across each chip's VCC/GND; all /OE pins to GND.
Limits: one NC switch per endpoint. MIN ends (34/36/39) need external 10k pull-ups to 3.3V; MAX ends (18/19/23) use the internal pull-ups.

GPIO	Function
26 / 25	neck-left STEP / DIR -> 74AHCT125 U1 -> DM556TE PUL+ / DIR+
33 / 32	neck-right STEP / DIR -> 74AHCT125 U1 -> DM556TE PUL+ / DIR+
27 / 14	neck-pivot STEP / DIR -> 74AHCT125 U2 -> DM556TE PUL+ / DIR+
4 / 13	Upstream UART1 RX / TX -- to j4_receiver: RX from TTGO GPIO 21, TX to TTGO GPIO 22
22 / 35	Downstream UART2 TX / RX -- to j4_stepper_eyes: TX to eyes GPIO 4, RX from eyes GPIO 13
34	neck-left MIN limit (input-only, external 10k pull-up)
36	neck-right MIN limit (input-only, external 10k pull-up)
39	neck-pivot MIN limit (input-only, external 10k pull-up)
18	neck-left MAX limit (internal pull-up)
19	neck-right MAX limit (internal pull-up)
23	neck-pivot MAX limit (internal pull-up)

j4_stepper_eyes -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM

LEFT rail			RIGHT rail		
			+-----+		
limit ER MIN --	36			3V3	3.3V (limit pull-ups)
limit EL MAX --	39			22	(free / onboard LED)
reset --	EN			19	(free)
limit EL MIN --	34			23	(free)
limit ER MAX --	35			18	(free)
DIR eye-R --	32			5	(free)
STEP eye-R --	33			17	TMC2209 UART TX
DIR eye-L --	25			16	TMC2209 UART RX
STEP eye-L --	26			4	UART1 RX <- neck 22
(free) --	27			0	(free / BOOT)
(free) --	14			2	(free)
(free) --	12			15	(free)
ground --	GND			13	UART1 TX -> neck 35
			+---[USB-C]---+		

Both TMC2209 share the UART bus (17/16); addresses 0b00 (eye-L) / 0b01 (eye-R).
Limits: one NC switch per endpoint, each with a 10k pull-up to 3.3V
(input-only pins have no internal pull-up).

GPIO	Function
26 / 25	eye-left STEP / DIR
33 / 32	eye-right STEP / DIR
17 / 16	TMC2209 UART TX / RX (shared PDN_UART bus)
4 / 13	Upstream UART1 RX / TX (to neck board)
34	eye-left MIN limit (input-only, external 10k pull-up)
39	eye-left MAX limit (input-only, external 10k pull-up)
36	eye-right MIN limit (input-only, external 10k pull-up)
35	eye-right MAX limit (input-only, external 10k pull-up)

j4_talk -- pin diagram

Teensy 4.1 + Audio Shield Rev D, component side up, USB at the TOP

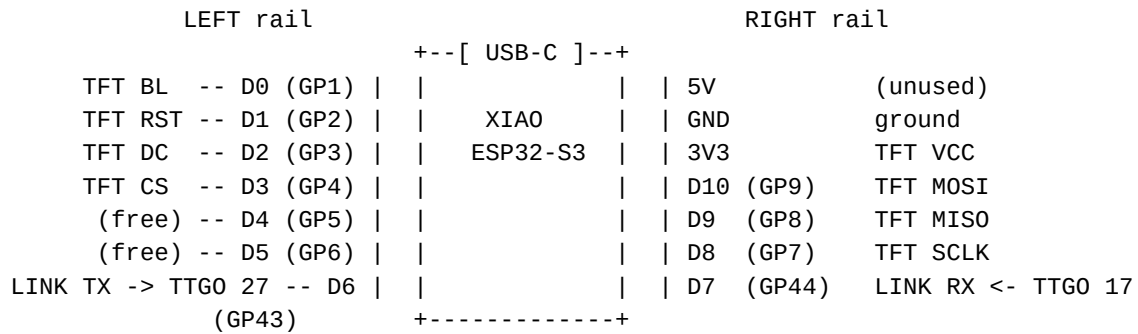
+===== [USB] =====+			
	GND	- GND	5V - 5V
	0	- 0	GND - GND
	1	- 1	3.3V - 3.3V
front_LED_01 (T5)	2	*- 2	23 - (audio I2S)
front_LED_02 (T4)	3	*- 3	22 -* front_LED_09 (T4)
front_LED_03 (T3)	4	*- 4	21 - (audio I2S)
front_LED_04 (T2)	5	*- 5	20 - (audio I2S)
front_LED_05 (T1)	6	*- 6	19 - (audio I2C)
(audio)	7	- 7	18 - (audio I2C)
(audio)	8	- 8	17 -* front_LED_08 (T3)
front_LED_06 (T1)	9	*- 9	16 -* front_LED_07 (T2)
side_LED_07 (T5)	10	*- 10	15 -* side_LED_01 (T2)
side_LED_08 (T5)	11	*- 11	14 -* front_LED_10 (T5)
	12	- 12	13 - (LED_BUILTIN)
Serial6 TX	24	*- 24	41 -
Serial6 RX	25	*- 25	40 -
	26	- 26	39 -
	27	- 27	38 -
side_LED_02 (T2)	28	*- 28	37 -* side_LED_06 (T4)
side_LED_03 (T3)	29	*- 29	36 -* side_LED_05 (T4)
	30	- 30	35 -
	31	- 31	34 -
	32	- 32	(bottom: SD card + SMD pads)
side_LED_04 (T3)	33	*- 33	
+=====+			

Serial6: 24 (TX) -> j4_receiver GPIO 2 ; 25 (RX) <- j4_receiver GPIO 17
Audio Shield Rev D uses 7, 8, 20, 21, 23 (I2S) + 18, 19 (I2C) + built-in SD.
front_LED_01..10 = front row left to right (pins 2,3,4,5,6,9,16,17,22,14);
front_05/06 light first. side_LED_01..08 = down the side, top to bottom
(pins 15,28,29,33,36,37,10,11).

Teensy Pin	Function
2	front_LED_01 (tier 5)
3	front_LED_02 (tier 4)
4	front_LED_03 (tier 3)
5	front_LED_04 (tier 2)
6	front_LED_05 (tier 1, innermost)
9	front_LED_06 (tier 1, innermost)
10	side_LED_07 (tier 5)
11	side_LED_08 (tier 5)
14	front_LED_10 (tier 5)
15	side_LED_01 (tier 2)
16	front_LED_07 (tier 2)
17	front_LED_08 (tier 3)
22	front_LED_09 (tier 4)
24	Serial6 TX to j4_receiver (receiver GPIO 2)
25	Serial6 RX from j4_receiver (receiver GPIO 17)
28	side_LED_02 (tier 2)
29	side_LED_03 (tier 3)
33	side_LED_04 (tier 3)
36	side_LED_05 (tier 4)
37	side_LED_06 (tier 4)

j4_display_left -- pin diagram

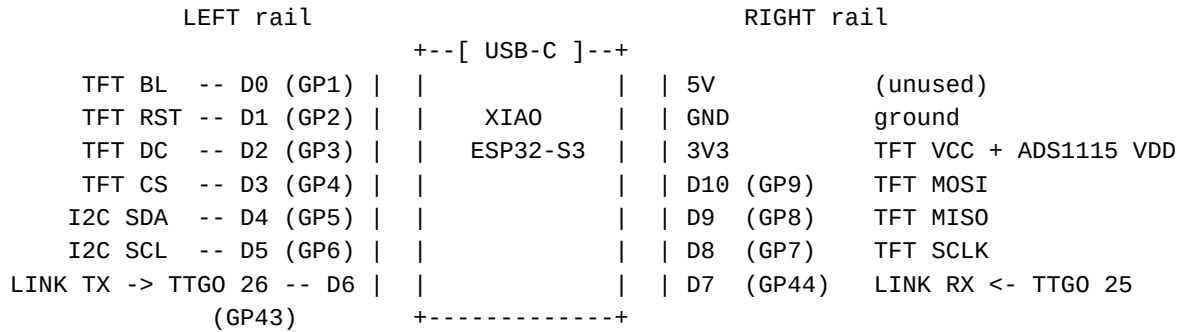
SeeedStudio XIAO ESP32S3, component side up, USB-C at the TOP



XIAO Pin (GPIO)	Function
3V3 (n/a)	Display VCC (3.3V power)
GND (n/a)	Display GND, shared with TTGO GND
D0 (GPIO1)	TFT backlight (HIGH = on)
D1 (GPIO2)	TFT RST
D2 (GPIO3)	TFT DC
D3 (GPIO4)	TFT CS
D4 (GPIO5)	Free (I2C SDA if needed later)
D5 (GPIO6)	Free (I2C SCL if needed later)
D6 (GPIO43)	UART TX to TTGO GPIO27
D7 (GPIO44)	UART RX from TTGO GPIO17
D8 (GPIO7)	TFT SCLK
D9 (GPIO8)	TFT MISO
D10 (GPIO9)	TFT MOSI

j4_display_right -- pin diagram

SeedStudio XIAO ESP32S3, component side up, USB-C at the TOP



D4/D5 are the XIAO's native I2C pins -- the ADS1115 hangs there (addr 0x48).
The TTGO->XIAO direction (D7) is wired but unused; reserved for later.

XIAO Pin (GPIO)	Function
3V3 (n/a)	Display VCC + ADS1115 VDD (3.3V power)
GND (n/a)	Display/ADS1115 GND, shared with TTGO GND
D0 (GPIO1)	TFT backlight (HIGH = on)
D1 (GPIO2)	TFT RST
D2 (GPIO3)	TFT DC
D3 (GPIO4)	TFT CS
D4 (GPIO5)	I2C SDA (ADS1115)
D5 (GPIO6)	I2C SCL (ADS1115)
D6 (GPIO43)	UART TX to TTGO GPIO26
D7 (GPIO44)	UART RX from TTGO GPIO25 (reserved, unused)
D8 (GPIO7)	TFT SCLK
D9 (GPIO8)	TFT MISO
D10 (GPIO9)	TFT MOSI